Acoustic simulation documentation

...

2026

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1 the pde

derivation at the wikipedia (TODO: provide link)
https://en.wikipedia.org/wiki/Acoustic_wave_equation

$$\frac{\partial^2 p(\boldsymbol{x},t)}{\partial t^2} = a^2 \Delta p(\boldsymbol{x},t) = a^2 \left(\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2} \right) \tag{1.1}$$

here Δ is the Laplace operator

2 the boundary conditions

Homogeneous Dirichlet: $p(\boldsymbol{x},t)|_{\boldsymbol{x}\in S_d} = 0$
Homogeneous Neumann: $\nabla_{\boldsymbol{x}} p(\boldsymbol{x},t) \cdot \boldsymbol{n}_n|_{\boldsymbol{x}\in S_n} = 0$
Inhomogeneous Neumann (Transducer): $\nabla_{\boldsymbol{x}} p(\boldsymbol{x},t) \cdot \boldsymbol{n}_t|_{\boldsymbol{x}\in S_t} = \rho\omega^2 d \sin(\omega t)$
(TODO: is this correct for the transducer?????) (TODO: add derivation later)
Where $\boldsymbol{n}_*$ is the normal direction of the boundary
adn S_* is the set of given boundary points
ρ is the fluid density
ω is the excitations angular frequency from the transducer
d is the displacement of the transducer

3 discretization

notations:

$$\frac{\partial f(\boldsymbol{x},t)}{\partial t} = \dot{f} \tag{3.1}$$

$$\frac{\partial g(x,t)}{\partial x} = g' \tag{3.2}$$

$$\tag{3.3}$$

for a uniform grid of:
$(t_0, t_1, ..., t_n, ..., t_N)$
$(x_0, x_1, ..., x_j, ..., x_J)$
$(y_0, y_1, ..., y_j, ..., y_J)$
the following notation is adapted: $p(x_j, y_i, t_n) = p^n_{j,i}$, and for 1D: $p(x_j, t_n) = p^n_j$
here x_0, y_0, x_J, y_J, are at the boundary and $x_J - x_0 = y_J - y_0 = L$, the length of the square simulation domain
$\Delta t = t_{n+1} - t_n$
$\Delta x = x_{j+1} - x_j = \Delta y = y_{i+1} - y_i$

3.1 CFL

$$\sigma = \frac{a\Delta t}{\Delta x} < 1 \tag{3.4}$$

$$\Delta t \text{ proportional } \Delta x \Rightarrow O(\Delta t) = O(\Delta x) \tag{3.5}$$

3.2 pde

the central difference method is used, that can be seen at 3.6

$$\left.\frac{\partial^2 f}{\partial x^2}\right|_{x=x_j} \approx \frac{f_{j+1} - 2f_j + f_{j-1}}{\Delta x^2} + O(\Delta x^2) \tag{3.6}$$

$$\frac{p^{n+1}_{j,i} - 2p^n_{j,i} + p^{n-1}_{j,i}}{\Delta t^2} = a^2 \left(\frac{p^n_{j+1,i} - 2p^n_{j,i} + p^n_{j-1,i}}{\Delta x^2} + \frac{p^n_{j,i+1} - 2p^n_{j,i} + p^n_{j,i-1}}{\Delta y^2} \right) + \tag{3.7}$$

$$p^{n+1}_{j,i} = 2p^n_{j,i} - p^{n-1}_{j,i} + \frac{a^2\Delta t^2}{\Delta x^2} \left(p^n_{j+1,i} + p^n_{j-1,i} + p^n_{j,i+1} + p^n_{j,i-1} - 4p^n_{j,i} \right) \tag{3.8}$$

3.3 bc

3.3.1 Homogeneous Dirichlet in 1D

$$p_{j_d}^n = 0 \tag{3.9}$$

3.3.2 Homogeneous Neumann in 1D, second order

$$\begin{cases} \ddot{p}(x,t) = a^2 p''(x,t) & |x \in (0,L) \\ p'(0,t) = 0 = p_0'^n & \\ p'(L,t) = p_J'^n = \rho d\omega^2 \sin(\omega t) & |x_J = L \end{cases} \tag{3.10}$$

$$\begin{aligned} \frac{p_1^n - p_0^n}{\Delta x} = p_{1/2}'^n + O(\Delta x^2) \overset{3.12}{=} p_0'^n - \frac{\frac{\Delta x}{2}}{1!} p_{1/2}''^n + O(\Delta x^2) \overset{3.13}{=} p_0'^n - \frac{\Delta x}{2} p_1''^n + O(\Delta x^2) \overset{3.14}{=} \\ \overset{3.14}{=} p_0'^n - \frac{\Delta x}{2} p_1''^{n-1} + O(\Delta x^2) \overset{3.15}{=} p_0'^n - \frac{\Delta x}{2c^2\Delta t^2}\left(p_1^n - 2p_1^{n-1} + p_1^{n-2}\right) + O(\Delta x^2) \end{aligned} \tag{3.11}$$

$$p_0'^n = p_{1/2}'^n + \frac{\frac{\Delta x}{2}}{1!} p_{1/2}''^n + O(\Delta x^2) \tag{3.12}$$

$$p_{1/2}''^n = p_1''^n - \frac{\frac{\Delta x}{2}}{1!} p_1'''^n + O(\Delta x^2) = p_1''^n + O(\Delta x) \tag{3.13}$$

$$p_1''^n = p_1''^{n-1} + O(\Delta t) \overset{CFL}{=} p_1''^{n-1} + O(\Delta x) \tag{3.14}$$

from the equation of motion

$$p_1''^{n-1} = \frac{1}{c^2}\ddot{p}_1^{n-1} = \frac{1}{c^2\Delta t^2}\left(p_1^n - 2p_1^{n-1} + p_1^{n-2}\right) + O(\Delta t^2) \tag{3.15}$$

$$\begin{aligned} p_0^n = p_1^n - \Delta x p_0'^n + \frac{\Delta x^2}{2a^2\Delta t^2}\left(p_1^n - 2p_1^{n-1} + p_1^{n-2}\right) + O(\Delta x^3) = \\ = p_1^n + \frac{\Delta x^2}{2a^2\Delta t^2}\left(p_1^n - 2p_1^{n-1} + p_1^{n-2}\right) + O(\Delta x^3) \end{aligned} \tag{3.16}$$

3.3.3 Inhomogeneous Neuann, Transducer in 1D, second order

similarily as the previous section

$$p_J^n = p_{J-1}^n + \Delta x p_J'^n - \frac{\Delta x^2}{2a^2\Delta t^2}\left(p_{J-1}^n - 2p_{J-1}^{n-1} + p_{J-1}^{n-2}\right) + O(\Delta x^3) \tag{3.17}$$

$$p_J^n = p_{J-1}^n + \Delta x \rho d\omega^2 \sin(\omega t_n) - \frac{\Delta x^2}{2a^2\Delta t^2}\left(p_{J-1}^n - 2p_{J-1}^{n-1} + p_{J-1}^{n-2}\right) + O(\Delta x^3) \tag{3.18}$$

$t_n = n\Delta t$

3.3.4 Impedance or Sommerfeld bc, in 1D, first order

$\frac{\partial p(x,t)}{\partial t} + Za\frac{\partial p(x,t)}{\partial n} = 0$
Note: if the normal of the boundary is (-1) (or in 2D $(-1,0)$) then $n = -x$
a is the speed of sound
Z is the impedance
$Z = 0 \Rightarrow$ Dirichlet (reflection with same phase)
$Z = 1 \Rightarrow$ Sommerfeld (no reflection, all is absorbed)

$Z \to \infty \Rightarrow$ Neumann (phase inverting reflection)
for $n = -x$, with Explicit Euler

$$\frac{p_0^n - p_0^{n-1}}{\Delta t} = Za\frac{p_1^{n-1} - p_0^{n-1}}{\Delta x} \tag{3.19}$$

$$p_0^n = p_0^{n-1} + \frac{Za}{\Delta x}\left(p_1^{n-1} - p_0^{n-1}\right) \tag{3.20}$$

for $n = x$, with Explicit Euler

$$\frac{p_J^n - p_J^{n-1}}{\Delta t} = -Za\frac{p_J^{n-1} - p_{J-1}^{n-1}}{\Delta x} \tag{3.21}$$

$$p_J^n = p_J^{n-1} - \frac{Za}{\Delta x}\left(p_J^{n-1} - p_{J-1}^{n-1}\right) \tag{3.22}$$